EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

REVISION

GS-1 (DYNAMIC)

MODULE - 14

ECOLOGY & ENVIRONMENT-3

What do know about the Central Groundwater Authority?

- It was formed under the Environment Protection Act, it is under the Ministry of Jal Shakti, Department of Water Resources, River Development and Ganga rejuvenation.
- It is established under Section 3(3) of the Environment Protection Act, 1986 to regulate and control development and management of groundwater resources in the country.
- It exercises powers under Section 5 of the EPA 1986, for issuing directions and taking such measures in respect of all the matters referred to it under various provisions of the EPA.
- It can resort to penal provisions contained in Sections 15 to 21 of the said Act. It exercises powers under Section 4 of the EPA, 1986 for the appointment of officers.

What is a pollution market (ETS)? What are the results of Gujarat study over a period of five years?

- Pollution market is an Emissions Trading Scheme or cap-and-trade system, which are in turn allocated tradable emission permits. In 2019, Gujarat set up the world's first such market for particulate matter emissions in Surat.
- The scheme set an overall cap on the mass of particulates that could be collectively emitted by 318 large coal using plants who were either allocated or allowed to buy permits.
- As per a study conducted after over five years, the scheme reduced particulate emissions by 20 to 30% and pollution abatement cost by over 10%. Compliance with environmental laws rose to 99% among participating plants.

What is the present status of the Carbon Markets in India?

- India has notified the National Designated Authority for the Implementation of the Article 6 of the Paris Agreement (NDAIAPA) on 30th May, 2022 with Secretary MoEFCC as the chairperson. NDAIAPA Committee has updated and finalized the list of 14 activities under GHG mitigation activities, alternate materials and removal activities.
- The Government of India announced the Carbon Credit Trading Scheme (CCTS) on June 28, 2023, under the Energy Conservation Act, 2001 (Amendment, 2022), with the aim of establishing India's carbon market.
- National Steering Committee for Indian Carbon Market (NSC-ICM) has been formed under CCTS to oversee the Indian Carbon Market (ICM). The NSCICM is chaired by the Secretary of the Ministry of Power and co-chaired by the Secretary of the Ministry of Environment, Forest, and Climate Change. It has finalized 9 sectors under compliance mechanism under CCTS.
- The committee provides recommendations to the Bureau of Energy Efficiency (BEE) on the procedures, rules, and regulations for the ICM. It also recommends specific greenhouse gas (GHG) emission targets for obligated entities.

What are the recent guidelines for the Co-firing with the biomass pellets for TPPs?

- The Union Ministry of Power has issued a new comprehensive policy mandating the co-firing of biomass pellets and MSW-derived torrefied charcoal in coal-based thermal power plants.
- The policy directs all coal-based thermal power plants in Delhi-National Capital Region (NCR) to use a 5 per cent (by weight) blend of biomass pellets and an additional 2 per cent blend of either biomass pellets or torrefied charcoal made from MSW, along with coal, with effect from financial year 2025-26. Other thermal power plants outside NCR will have to co-fire 5 per cent (by weight) of biomass pellets or torrefied MSW charcoal from 2025-26.
- It specifies that for TPPs located in NCR and adjoining areas, a minimum of 50 per cent of the raw material in pellets must consist of stubble, straw or rice paddy crop residue sourced exclusively from NCR and adjoining regions.

What do you know about the National Mission on Natural Farming (NMNF)?

- Union Cabinet approved the NMNF as a standalone Centrally Sponsored Scheme under the Ministry of Agriculture and Farmers Welfare. The scheme has a total outlay of Rs. 2,481 crore (Government of India share - Rs.1,584 crore, State share – Rs. 897 crore) till the 15th Finance Commission (2025-26).
- It is a chemical free farming which involves local livestock integrated natural farming methods, diversified crop systems, etc. It follows local agro-ecological principles rooted in local knowledge.
- In the two years, till March 2026, NMNF will be implemented in 15,000 clusters in Gram Panchayats, which are willing and reach 1 crore farmers and initiate Natural Farming in 7.5 lakh hectares area.
- Preference will be given to areas having prevalence of practising NF farmers, SRLM / PACS / FPOs, etc. Need-based 10,000 Bio-input Resource Centres (BRCs) will be set-up to provide easy availability and accessibility to ready-to-use NF inputs for farmers.
- Around 2,000 NF Model Demonstration Farms will be established at Krishi Vigyan Kendras (KVKs), Agricultural Universities (AUs) and farmers' fields, and they shall be supported by experienced and trained Farmer Master Trainers.

What are the efforts of the Government to promote organic agriculture?

The ICAR has developed improved and efficient strains of bio-fertilizers specific to different crops and soil types under the Network project on 'Soil Biodiversity-Bio-fertilizers'.

- Government is implementing Market Development Assistance (MDA) at Rs. 1500/Metric Tonne to promote organic fertilizers, i.e., Fermented Organic Manure/ Liquid Fermented Organic Manure/Phosphate Rich Organic Manure produced at plants under GOBARdhan scheme of Ministry of Jal Shakti, Department of Drinking Water and Sanitation.
- For organic farming under PKVY, assistance of Rs. 31,500 per hectare for a period of three years is provided for promotion of organic farming. Out of this, assistance of Rs. 15,000 per hectare for a period of three years is provided to farmers through DBT.
- There is also PM Programme for Restoration, Awareness, Nourishment and Amelioration of Mother Earth (PM-PRANAM), which incentivizes States and Union Territories to promote alternative fertilizers. Under this programme, 50% of subsidy savings will be passed on as a grant to the state that reduces chemical fertilizers.
- National Center of Organic and Natural Farming (NCONF) under the Ministry of Agriculture undertakes training programmes and online awareness campaign with regard to organic and natural farming.

List out some achievements on the Renewable Energy front in recent years?

- The Government launched the PM Surya Ghar Muft Bijli Yojana targeting 1 crore households. This provides up to 300 units free electricity monthly and offers subsidies ranging from Rs. 30,000 to Rs. 78,000 per household.
- MNRE continues implementation of the National Green Hydrogen Mission with an outlay of Rs. 19,744 cr.
- The Ministry of Environment has notified the E-Waste (Management) Rules, 2022, effective from April 1, 2023, which apply to solar photovoltaic modules, panels, or cells.
- Union Cabinet approved a Rs. 7,453 crore Viability Gap Funding (VGF) Scheme to set up India's first offshore wind energy projects. The Scheme includes Rs. 6,853 crore for 1 GW of offshore wind capacity (500 MW each off the coasts of Gujarat and Tamil Nadu) and Rs. 600 crore for port upgrades to support logistics for these projects.
- Indian Renewable Energy Development Agency (IRADA) secured International Credit Rating BBB-long-term and A-3 short-term from S&P Global with a stable outlook.
- Solar Energy Corporation of India limited (SECI) attained Navratna status.
- Singareni Collieries Company Limited (SCCL) commissioned a 20 kW, pilot geothermal power plant in Manuguru, Telangana.
- The Central Financial Assistance for Pellet manufacturing plant stands at Rs. 21 lakh per MTPH, maximum Rs. 105 lakhs per project, while for torrefied pellet plants it is almost double.

What do you understand by Social Cost of Carbon (SCC)?

- Social Cost of Carbon (SCC) is a concept used by economists and policymakers to quantify and monetize the long-term economic damages associated with an incremental increase in carbon dioxide emissions in a given year. This includes a wide range of impacts, such as:
  - ✓ Changes in net agricultural productivity.
  - ✓ Human health.
  - ✓ Property damages from increased flood risk.
  - ✓ Changes in energy system costs, among others.
- It is used to estimate the value of damages avoided for a small emission reduction. In other words, it talks about damages incurred by an additional ton of carbon emissions.
- It provides a way to evaluate the economic benefits of policies that reduces greenhouse gas emissions by comparing them to the policy costs.

Social Cost of Carbon plays a crucial role in the development of climate policy and environmental regulation by helping to inform cost-effective analyses of climate actions.

What are the environmental consequences of various pollutants released by coal based thermal power plants?

- When coal is burned by power plants, it releases various pollutants into the atmosphere. Among these pollutants are sulfur dioxide (SO_2), nitrogen oxides (NO_x).
- SO_2 contribute to acid rain and respiratory problems, while NO_X plays a significant role in the formation of ground level ozone and smog also contributing to respiratory issues.
- Indian coal has high ash content, often ranging from 30 to 45% and this poses challenges for combustion and pollution control in power plants.
- Ash contains a variety of heavy metals, including arsenic, lead and mercury. If not properly managed, they can contaminate water sources and soil.

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